LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 4

Course Code: CSE276
Course Title: ARTIFICIAL INTELLIGENCE FOUNDATIONS
Course Outcome: CO4
Paper: Sample Multiple Choice Questions
Maximum Marks: 40 x 1 = 40

Unit: 4
Unit Title: Evolution of Modern AI and Foundation Models
Unit Mapped RBT: L3 (Apply)
Total Questions: 40
Suggested Duration: 60 Minutes

Answer Grid (to be filled by the student)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1		Q11		Q21		Q31	
Q2		Q12		Q22		Q32	
Q3		Q13		Q23		Q33	
Q4		Q14		Q24		Q34	
Q5		Q15		Q25		Q35	
Q6		Q16		Q26		Q36	
Q7		Q17		Q27		Q37	
Q8		Q18		Q28		Q38	
Q9		Q19		Q29		Q39	
Q10		Q20		Q30		Q40	

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the deep learning architecture designed for image data.

A) The convolutional neural network
B) The recurrent neural network
C) The decision tree ensemble
D) The production rule system

Q2. Identify the type of machine learning in which the system learns from a reward received after each action.

A) Supervised learning
B) Reinforcement learning
C) Unsupervised learning
D) Semi-supervised learning

Q3. State what Edge Artificial Intelligence means.

A) Running the model on a remote cluster of servers
B) Training the model on data drawn from many devices
C) Running the model on the device that collects the data
D) Running the model only at the edge of a network cable

Q4. Name the technique that reduces the numeric precision used to store a model's weights.

A) Pruning
B) Fine-tuning
C) Augmentation
D) Quantisation

Q5. Identify what a foundation model is pre-trained on.

A) A small dataset from one specialised domain
B) A very large and broad body of general data
C) A labelled dataset for one narrow task
D) The outputs of an earlier expert system

Q6. State the three types of machine learning named on the syllabus.

A) Symbolic, statistical and hybrid learning
B) Shallow, deep and very deep learning
C) Local, edge and cloud based learning
D) Supervised, unsupervised and reinforcement learning

Q7. Explain why symbolic Artificial Intelligence was described as brittle.

A) It produced no answer for a case its rules did not cover
B) It ran too slowly to be used on real problems
C) It required an extremely large training dataset
D) It could not represent any form of knowledge

Q8. Distinguish training from inference in the life of a model.

A) Inference fits the parameters, training applies them
B) Both of the two fit the parameters of the model
C) Training fits the parameters, inference applies them
D) Both of the two apply the parameters to new data

Q9. Describe what an agentic Artificial Intelligence system does beyond answering a question.

A) It answers the same question much more quickly
B) It stores every question it has ever been asked
C) It plans steps, calls tools and revises its own plan
D) It refuses any question outside its training data

Q10. Interpret the following as a type of machine learning: a system is given customer records each carrying a known outcome of whether that customer left.

A) Supervised learning
B) Unsupervised learning
C) Reinforcement learning
D) Semi-supervised learning

Q11. Predict the effect on latency and on privacy of running a model on the device rather than in the cloud.

A) Latency rises and the raw data must leave the device
B) Latency falls but the raw data must leave the device
C) Latency rises but the raw data stays on the device
D) Latency falls and the raw data need not leave the device

Q12. Recognize the deep learning application described: a system converts spoken audio into written text.

A) Image classification
B) Speech recognition
C) Object detection
D) Recommendation

Q13. Explain why deep learning became practical only comparatively recently.

A) The underlying mathematics was only just discovered
B) Symbolic methods were legally prohibited
C) Neural networks were invented in the last decade
D) Large datasets and fast parallel hardware became available

Q14. Distinguish a foundation model from a model trained for one particular task.

A) A task model is broad and then adapted to one task
B) Both of the two are trained for one task only
C) A foundation model is broad and then adapted to tasks
D) Both of the two are trained on general data only

Q15. Explain what it means for an Artificial Intelligence model to be multimodal.

A) It can be deployed on more than one kind of device
B) It accepts more than one kind of input, such as text and images
C) It was trained by more than one organisation
D) It produces more than one answer per question

Q16. Describe the constraint that distinguishes Tiny Artificial Intelligence from Edge Artificial Intelligence.

A) Tiny AI targets microcontrollers with very little memory and power
B) Tiny AI targets servers with very large memory
C) Tiny AI is trained on a much smaller dataset
D) Tiny AI produces much shorter answers

Q17. Predict what a statistical model does when it meets an input quite unlike anything in its training data.

A) It reports that the input is outside its experience
B) It still produces an output, with no warning of the difference
C) It refuses to produce any output whatsoever
D) It automatically retrains itself on the new input

Q18. Summarize what statistical Artificial Intelligence supplied that the symbolic approach could not.

A) Rules inferred from examples rather than stated by hand
B) Guaranteed correctness for every possible case
C) An explanation of every decision it produces
D) Freedom from any need for training data

Q19. Explain why a recurrent architecture suits data that arrives as a sequence.

A) It processes every position independently
B) It requires the sequence to be of fixed length
C) It converts the sequence into a single image
D) It carries information forward from earlier positions

Q20. Distinguish pruning from quantisation as ways of making a model smaller.

A) Quantisation removes weights, pruning stores them in fewer bits
B) Both of the two remove weights from the network
C) Pruning removes weights, quantisation stores them in fewer bits

D) Both of the two reduce the precision of the weights

Q21. Solve the following: a model holding 3 billion parameters is stored at 2 bytes for each parameter. Determine the memory its weights occupy.

A) 6 GB
B) 3 GB
C) 1.5 GB
D) 12 GB

Q22. Compute the memory saved by quantising a model from 2 bytes per parameter to 1 byte.

A) A quarter of the original
B) Three quarters of the original
C) Half of the original
D) None of the original

Q23. Apply the three types of machine learning to a task in which fifty thousand records carry no labels and the request is to divide them into groups.

A) Supervised learning
B) Unsupervised learning
C) Reinforcement learning
D) Transfer learning

Q24. Use the distinction between pre-training and fine-tuning to determine which of the two is repeated for each new domain.

A) Pre-training
B) Both of the two equally
C) Neither of the two
D) Fine-tuning

Q25. Examine the following and determine the cause: a vision model trained on daylight photographs performs badly on images taken at night.

A) The model has too many parameters to work
B) The images were stored in the wrong file format
C) The training data did not cover that condition
D) The model was quantised before deployment

Q26. Solve the following: a hosted model charges 0.0015 dollars for each query and 40,000 queries are made in a month. Determine the monthly cost.

A) 6 dollars
B) 600 dollars
C) 40 dollars
D) 60 dollars

Q27. Apply Edge Artificial Intelligence to a setting in which the network connection is frequently unavailable, and determine the benefit.

A) The system keeps working while the connection is down
B) The system trains itself faster while offline
C) The system needs a much larger model to run
D) The system stops needing any sensors at all

Q28. Compare a convolutional network with a recurrent network on the kind of data each is suited to.

A) Recurrent suits images, convolutional suits sequences
B) Convolutional suits images, recurrent suits sequences
C) Both of the two are designed for image data
D) Both of the two are designed for sequential data

Q29. Use transfer learning on a task holding only 500 labelled images and determine the usual procedure.

A) Train every layer of the network from the beginning
B) Discard the pre-trained network and use rules instead
C) Collect no further data and accept the poor result
D) Keep the pre-trained feature layers and retrain the last ones

Q30. Compute the memory occupied by 1.5 billion parameters stored at 4 bytes each.

A) 1.5 GB
B) 6 GB
C) 3 GB
D) 12 GB

Q31. Examine the following and determine the remedy: a general purpose model answers everyday questions well but handles the wording of medical reports poorly.

A) Increase the number of queries it is asked
B) Reduce the precision of its stored weights
C) Adapt it with domain data or retrieve from domain documents
D) Deploy it on a device rather than a server

Q32. Apply the definition of agentic Artificial Intelligence to determine which system qualifies.

A) One that plans steps, uses tools and revises its plan
B) One that answers a single question and stops
C) One that stores its answers in a database
D) One that runs entirely on a small device

Q33. Use the three types of machine learning to classify a system given labelled photographs of defective and sound welds.

A) Supervised learning
B) Unsupervised learning
C) Reinforcement learning
D) Agentic learning

Q34. Solve the following: a model stored at 4 bytes per parameter is quantised to 1 byte per parameter. Determine the resulting size as a percentage of the original.

A) 50 per cent
B) 75 per cent
C) 12.5 per cent
D) 25 per cent

Q35. Compare Edge Artificial Intelligence with cloud inference on the network bandwidth each one consumes.

A) The cloud sends little or nothing, edge sends every input
B) Edge sends little or nothing, the cloud sends every input
C) Both of the two send every input over the network
D) Neither of the two uses any network bandwidth

Q36. Examine the risk of fine-tuning a general model on a very small specialised dataset.

A) It becomes unable to run on any device
B) It loses the ability to accept text input
C) It may lose general ability while overfitting the small set
D) It requires the pre-training to be repeated

Q37. Apply the idea of distillation and determine what it produces.

A) A larger model built from several small ones
B) A model with its weights stored in fewer bits
C) A smaller model trained to imitate a larger one
D) A model with its least useful weights removed

Q38. Use the relation between model size and serving cost to determine the effect of halving the number of parameters.

A) The memory needed at inference is roughly doubled
B) The memory needed at inference is roughly halved
C) The memory needed at inference is unchanged
D) The memory needed at inference falls to zero

Q39. Demonstrate how a multimodal model is applied to a task that combines a photograph with a question written about it.

A) It answers from the text and ignores the photograph
B) It answers from the photograph and ignores the text
C) It answers the two inputs as two separate questions
D) It represents both inputs together and answers from the pair

Q40. Relate the volume of training data available to the choice between a deep network and a classical method.

A) Scarce data favours a classical method or transfer learning
B) Scarce data always favours a very deep network
C) The volume of data has no bearing on the choice
D) Abundant data always favours a classical method

----- End of Sample Question Set -----